

Midterm project

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Alphabet Inc. (Google) and Tesla were selected for this financial analysis. They are both pioneering American companies in technology and manufacturing sectors. While Google is a more established company, Tesla is catching up very quickly. A comparison between financial ratios of a technology company and a company that primarily sells products (with all that entails: manufacturing, distribution, expensive inventory, longer development-to-production cycles, etc.) could be interesting and demonstrate how financial metrics reflect real-world performance.

Google is an American technology company that was founded on September 4, 1998 and had its IPO in 2004. Google currently has the following spheres of business interests:

- artificial intelligence,
- online advertising,
- search engine technology,
- cloud computing,
- computer software,
- quantum computing,
- e-commerce,
- consumer electronics.

Google has the following products: email (Gmail), navigation (Waze & Maps), cloud computing, web browsing (Chrome), video sharing (YouTube), productivity (Workspace), operating systems (Android), cloud storage (Drive), language translation (Translate), photo storage (Photos), video calling (Meet), smart home (Nest), smartphones (Pixel), wearable technology (Pixel Watch & Fitbit), music streaming (YouTube Music), video on demand (YouTube TV), artificial intelligence (Google Assistant & Bard), machine learning APIs (TensorFlow), AI chips (TPU).

Tesla, Inc. is an American multinational automotive and clean energy company headquartered in Austin, Texas, which designs, develops software, manufactures and sells electric vehicles, stationary battery energy storage devices from home to grid-scale, solar panels and solar shingles, and related products and services. The company was founded in 2003 and became public in 2010. In 2008, the company began production of its first car model, the Roadster sports car, followed by

the Model S sedan in 2012, the Model X SUV in 2015, the Model 3 sedan in 2017, the Model Y crossover in 2020, the Tesla Semi truck in 2022 and the Cybertruck pickup truck in 2023. As of 2023, Tesla is the world's most valuable automaker. The Model 3 is the all-time bestselling plug-in electric car worldwide, and in June 2021 became the first electric car to sell 1 million units globally. In 2023, the Model Y was the best-selling vehicle, of any kind, globally. Tesla's production output in 2023 was 1,845,985 vehicles.

Data and data preparation

Financial data for the analysis was directly sourced from the U.S. Securities and Exchange Commission's EDGAR database using their API. The analysis focuses on five key financial ratios to assess both companies' profitability, liquidity and solvency: Current Ratio, Quick Ratio, Net Profit Margin, Return on Equity, and Debt-to-Equity.

Data cleaning and preparation

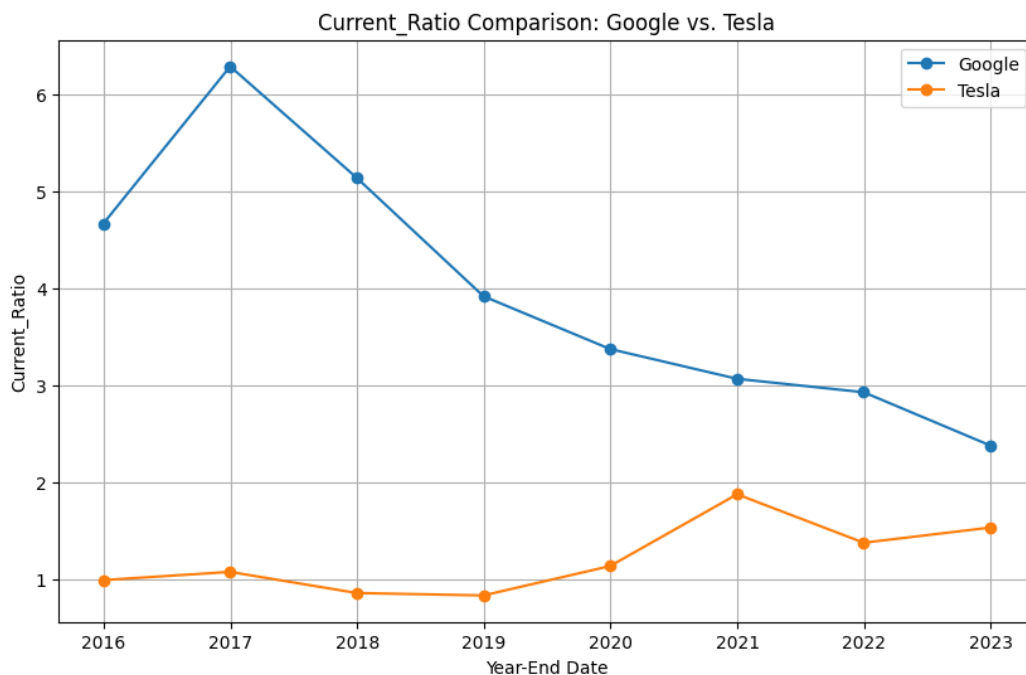
- The time frame for the data varied due to different age of the companies, so a decision was made to focus on the 2015 – 2022 time period.
- The following data preparation process was applied:
 - filtered the dataset to include only 10-K and 10-K/A reports
 - variables of interest were examined for missing values, and since there were none, there was no need to for any elimination/substitution methods.
 - there was an issue with multiple 10-K filings for the same fiscal year, which could introduce redundancies into the dataset. To resolve this, the data was filtered by the reporting period start and end date, then grouped and only the latest entry for each year was retained, ensuring that each fiscal year was represented by a single, unique data point in the resulting data frame.
 - Created an alternative key for Google's revenue, as for 2022 it was moved from 'Revenues' to 'Revenue from Contract with Customer excluding Assessed Tax'
 - Renamed the columns with the variables of interest for clarity and added a company tag.
 - Calculated ratios as new columns.
 - Removed any columns that were not directly related to the financial ratios in the final dataset.

- Combined all the ratios into a single dataframe, exported it to csv for report.
- Graphed the ratios

Liquidity Ratios

Current Ratio - Current Assets / Current Liabilities - measures the company's ability to meet its short-term obligations with its short-term assets.

	Current_Ratio_TSLA	Current_Ratio_GOOG
12/31/15	0.99	4.67
12/31/16	1.07	6.29
12/31/17	0.86	5.14
12/31/18	0.83	3.92
12/31/19	1.13	3.37
12/31/20	1.88	3.07
12/31/21	1.38	2.93
12/31/22	1.53	2.38



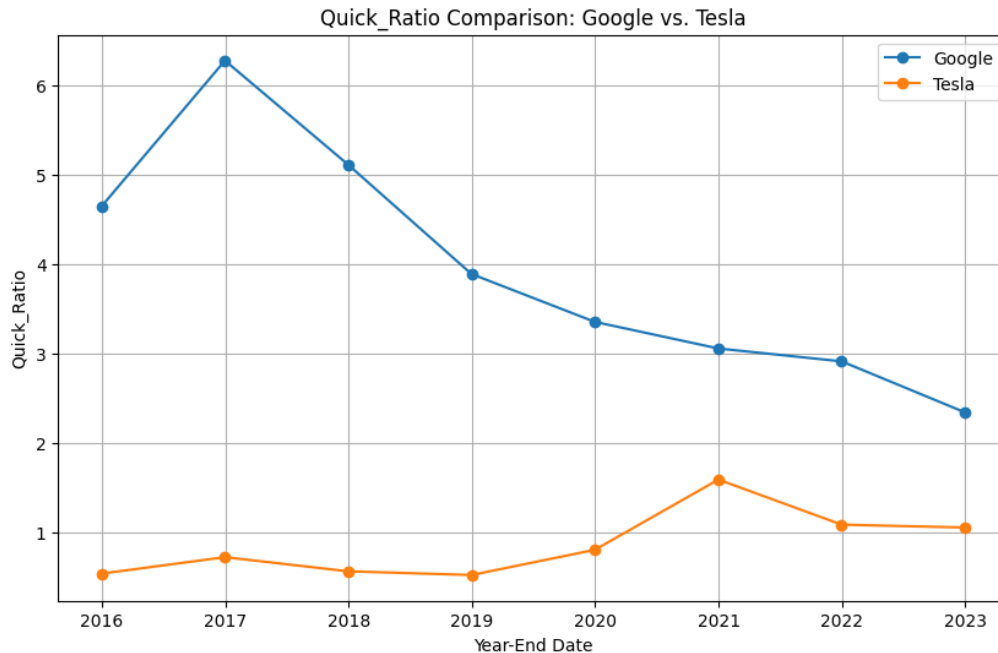
Google has consistently maintained a higher current ratio than Tesla. Google's current ratio shows a declining trend, but the ratio remains above 1, which generally indicates that the company has more than enough current assets to cover its current liabilities.

Tesla's current ratio, on the other hand, has been below Google's throughout the period, starting just above 1 in 2015. The ratio shows a slight upward trend from 2019 to 2022, suggesting an improving ability to cover its current liabilities. However, its current ratio has consistently stayed closer to 1, which may indicate tighter liquidity compared to Google.

A higher current ratio might suggest a more robust liquidity position, which can be a sign of financial health and stability. However, an excessively high current ratio (like the one Google had maintained through 2015 – 2019) could also mean that the company is not using its current assets efficiently or is not investing excess cash to generate more revenue.

Quick Ratio (Acid-test Ratio) - $(\text{Current Assets} - \text{Inventories}) / \text{Current Liabilities}$ – is similar to the current ratio but provides a more stringent measure of liquidity by excluding inventory (less liquid asset).

	Quick_Ratio_TSLA	Quick_Ratio_GOOG
12/31/15	0.54	4.64
12/31/16	0.72	6.27
12/31/17	0.56	5.11
12/31/18	0.52	3.89
12/31/19	0.80	3.35
12/31/20	1.59	3.05
12/31/21	1.08	2.91
12/31/22	1.05	2.34



Google begins with a quick ratio significantly higher than 1 in 2015. Over time, Google's quick ratio declines steadily, reaching closer to 1 by the end of 2022. Despite the decline, Google maintains a quick ratio above 1 throughout the period, which indicates that the company has sufficient liquid assets to cover its current liabilities.

Tesla's quick ratio is consistently lower than Google's for the entire period shown. It starts just above 0.5 in 2015, which indicates a less favorable liquidity position as the company might struggle to cover short-term liabilities without selling inventory. There is some fluctuation in Tesla's quick ratio, but by 2022, their quick ratio approaches 1, suggesting an improvement in its ability to cover its immediate obligations without relying on the sale of inventory. It should be noted that historically Tesla had undertaken significant efforts to boost end-of-quarter deliveries of its cars, therefore reducing inventory for the reporting period.

It is also interesting to compare these numbers to the industry median. According to <https://www.readyratios.com> median ratios for services industry (Google) in 2021 were: 1.47 current ratio, 1.24 quick ratio. For Manufacturing industry (Tesla) median current ratio was 2.88, and median quick ratio was 1.55. It's interesting but not surprising how much lower than then current is the quick ratio for manufacturing companies that have a lot of inventory. Google stays well above the industry benchmarks, while Tesla stays somewhat below.

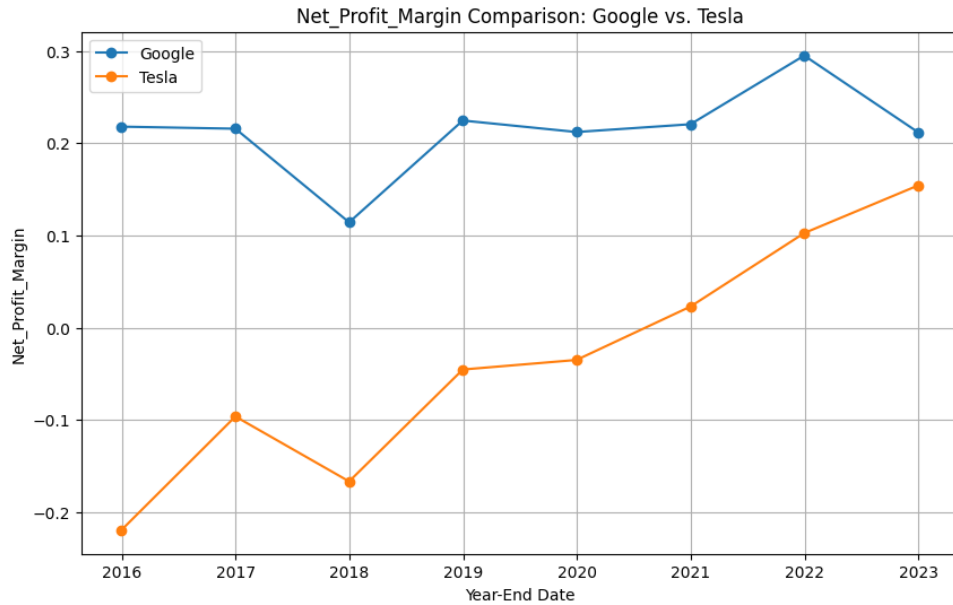
Overall, Google's liquidity is strong as indicated by its higher current and quick ratios over the years. Although both ratios have decreased over time, they remain at levels that suggest Google can easily meet its short-term obligations without the urgent sale of assets, which is indicative of financial stability. Overall, Google's liquidity ratios suggest a more conservative approach to liquidity management, maintaining a larger safety net of liquid assets.

Tesla, while showing an improving trend in both liquidity ratios, has consistently lower ratios compared to Google, implying tighter liquidity. The lower quick ratio suggests that a significant portion of Tesla's current assets might be tied up in inventory, which is not as readily convertible to cash. This could be reflective of the capital-intensive nature of Tesla's business in manufacturing. The combination of both ratios indicates that Tesla operates with a narrower liquidity margin, which might raise concerns about its ability to meet sudden short-term obligations without potentially disrupting operations.

Profitability Ratios

Net Profit Margin - Net Income / Revenue - Indicates how much profit a company makes for every dollar of its revenue.

	Net_Profit_Margin_TSLA	Net_Profit_Margin_GOOG
12/31/15	-22%	22%
12/31/16	-10%	22%
12/31/17	-17%	11%
12/31/18	-5%	22%
12/31/19	-4%	21%
12/31/20	2%	22%
12/31/21	10%	30%
12/31/22	15%	21%

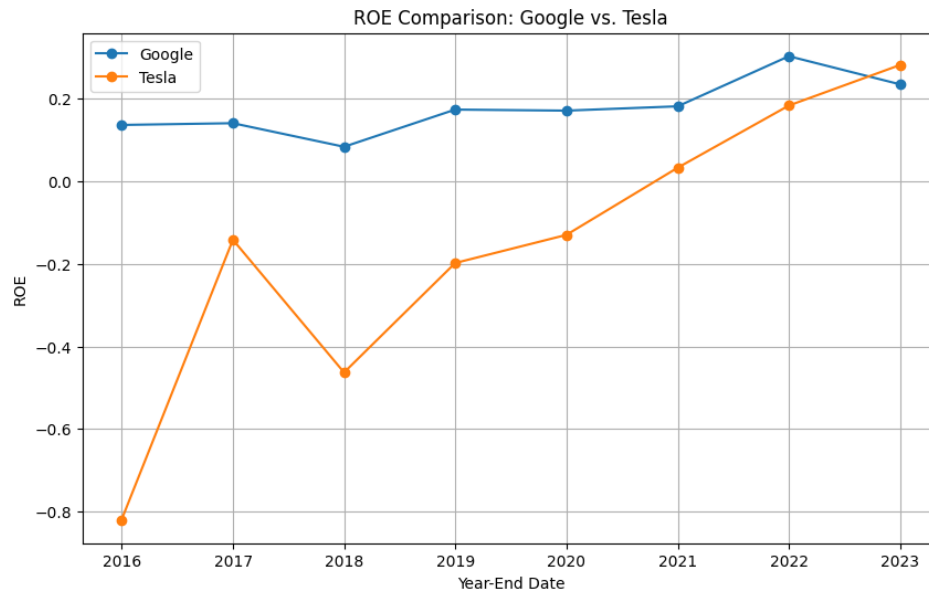


Google shows a relatively stable net profit margin over the years. The margin peaks in certain years, suggesting periods of particularly efficient cost management or higher revenue growth. Google's net profit margin remains positive throughout the period, reflecting consistent profitability.

Tesla's net profit margin starts negative in 2015 but over the years, company's margin improves steadily, crossing into positive territory and continuing to rise, showing that the company not only became profitable but also managed to increase its profitability over time. This upward trajectory could reflect Tesla's growth phase, improving operational efficiency, economies of scale, or increasing demand for its products (or all of the above).

Return on Equity (ROE) - $\text{Net Income} / \text{Shareholders' Equity}$ - measures the profitability of a corporation in relation to stockholders' equity.

	ROE_TSLA	ROE_GOOG
12/31/15	-82%	14%
12/31/16	-14%	14%
12/31/17	-46%	8%
12/31/18	-20%	17%
12/31/19	-13%	17%
12/31/20	3%	18%
12/31/21	18%	30%
12/31/22	28%	23%



Google's ROE remains positive throughout the period, showing some variability but generally hovering around a stable value. This indicates that Google has been consistently generating profit from its equity, which suggests efficient use of shareholder capital.

In contrast, Tesla's ROE starts off negative in 2015. However, from this low point, Tesla's ROE improves significantly over the years, eventually surpassing the breakeven point and becoming positive. This improvement suggests that Tesla has substantially increased its profitability in relation to shareholder equity, which could be a result of increased sales, improved operational efficiency, or successful new product launches.

According to <https://www.readyratios.com> median ROE for services industry in 2021 was -12.8% and -13% for manufacturing, so both companies are well above their industry benchmarks.

Google has demonstrated consistent profitability with a stable net profit margin and a positive ROE, reflecting a mature company that efficiently converts revenue into profit and effectively utilizes its equity to sustain growth. This consistency is often associated with a lower risk profile and may be more appealing to investors seeking steady returns.

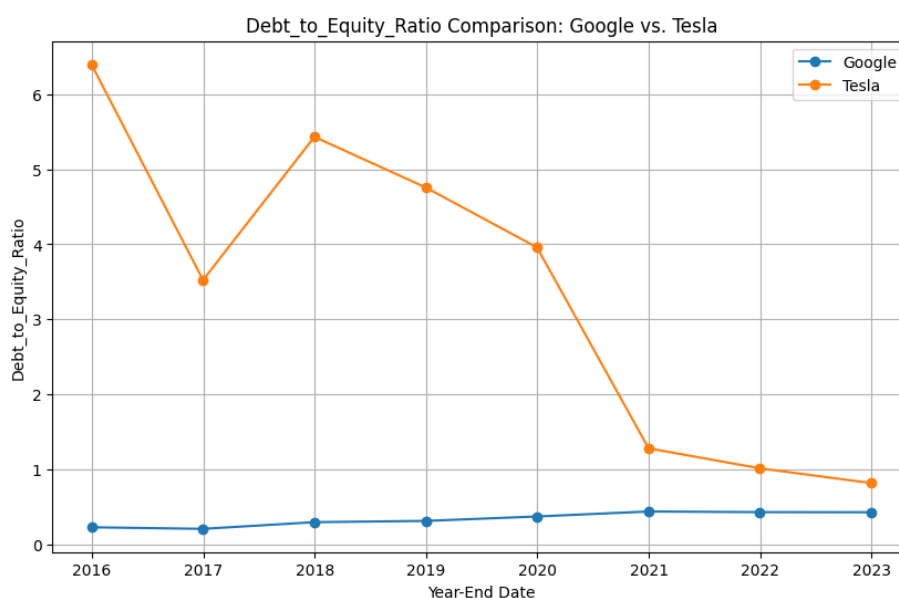
Tesla, conversely, has shown a remarkable improvement in both ratios over the observed period. This trend signifies Tesla's transition from a new company that struggled to generate shareholder returns to one that is steadily profitable.

Together, these profitability ratios tell a story of two companies at different life stages: Google as an established entity with solid financial health and Tesla as a dynamic, evolving company that has recently begun to realize its potential for profitability.

Solvency Ratio

Debt-to-Equity Ratio - Total Liabilities / Shareholders' Equity - gives an insight into the company's financial leverage and risk.

	Debt_to_Equity_Ratio_TSLA	Debt_to_Equity_Ratio_GOOG
12/31/15	6.40	0.23
12/31/16	3.52	0.20
12/31/17	5.43	0.29
12/31/18	4.76	0.31
12/31/19	3.96	0.37
12/31/20	1.28	0.44
12/31/21	1.01	0.43
12/31/22	0.82	0.43



Google's debt to equity ratio stays very low over the entire period, suggesting that the company has financed most of its operations through equity rather than debt. This can be a sign of financial stability and conservative financial management, as relying less on debt reduces the financial risk and interest expenses. However, it could also indicate that Google may not be taking full advantage of the potential growth opportunities that could be financed through debt.

Tesla's debt to equity ratio, on the other hand, starts very high in 2015, indicating that the company was heavily reliant on debt to finance its growth. Over time, this ratio dramatically decreases, suggesting that Tesla has been paying down debt or increasing its equity, or a combination of both,

thus reducing its financial leverage. This reduction could imply that Tesla is moving towards a more sustainable financial structure by relying less on debt, which could reduce the financial risk for equity holders.

According to <https://www.readyratios.com> median Debt to equity ratio in 2021 was 0.76 for services industry and 0.58 for manufacturing, so Tesla was above, while Google was below their industry benchmarks.

Conclusion

The five financial ratios — current ratio, quick ratio, net profit margin, return on equity (ROE), and debt to equity ratio — paint a comprehensive picture of Google as a mature, financially stable company with consistent profitability, strong liquidity, and low leverage, which may be more appealing to investors looking for lower risk and steady returns. Google's financials suggest a conservative management approach with a focus on sustaining growth while maintaining a strong balance sheet.

Tesla, by contrast, is portrayed through these ratios as a company that has experienced significant financial improvement and growth over the years. Initially, Tesla exhibited higher financial risk, as seen in its lower liquidity ratios and negative profitability metrics, along with a high debt to equity ratio. However, Tesla's positive trend in these financial metrics over time indicates a substantial turnaround, with improving profitability and a healthier financial leverage situation. This improvement may be particularly attractive to growth-focused investors who are willing to tolerate higher levels of risk for the potential of greater long-term gains, betting on Tesla's continued growth and innovation in its sector.

In summary, Google may fit the profile for investors seeking a more traditional, stable investment, while Tesla represents a more aggressive growth opportunity with a corresponding risk profile.